Alternative Risk-averse and Risk-taking Portfolio Constructions

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Abstract

This paper develops alternative portfolio constructions for the NAGMO Investment Thesis, which expresses a relative-value view through a long position in NVIDIA and short positions in Amazon and Google. It takes as its starting point the distinction between accounting balance and genuine market neutrality, the continuing presence of AI model risk, and the conditional dependence of nominal-rate critiques on the sign and magnitude of the U.S. inflation risk premium, ϕ. Rather than treating one set of weights as universally optimal, the paper specifies a family of risk-averse and risk-taking implementations. These include a constrained beta-neutral spread, a low-gross-exposure construction, a regime-aware defensive portfolio, a conviction-weighted relative-value portfolio, and a controlled gross-exposure portfolio. Each is expressed as an optimization or sizing rule, with explicit treatment of financing, borrow, transaction costs, factor exposures, technological scenarios, and uncertainty about ϕ. Numerical weights are deliberately not presented as empirically optimal: the source materials provide a stylized balance-sheet construction and an illustrative beta calibration, but not the return, covariance, financing, borrow, or model-risk data needed to estimate investable weights. The central recommendation is to select portfolio architecture by risk budget and regime, then estimate its weights from current, executable data.

The paper ends with "The End"

1 Purpose and design principles

The preceding review of the NAGMO Investment Thesis established five points that govern the constructions developed here.

First, the original NAGMO vector,

$$(a, b, c, d, e) = (1, -1, -1, 0, -0.42), \tag{1}$$

balances the stated market-capitalization coefficients:

$$5.03 - 2.40 - 2.21 - 0.42 = 0. \tag{2}$$

This is an accounting identity, not proof of zero market beta or factor neutrality. Second, the relative-value hypothesis—that NVIDIA's AI hardware position may outperform the cloud and service exposures represented by Amazon and Google—is a testable economic proposition, not an invariant law. Third, AI model risk persists across all portfolio architectures: changes in compute intensity, model efficiency, custom silicon, open-weight competition, and monetization can change the relative economics of the securities. Fourth, nominal beta, duration, discount-rate and carry critiques are conditional on the U.S. inflation risk premium, ϕ. Fifth, the value of a construction must be assessed after financing, borrow, transaction costs, liquidity, and tail risk.

This paper therefore does not declare a single portfolio to be the correct NAGMO portfolio. It defines a design space. A risk-averse construction seeks to preserve the relative-value thesis while limiting unwanted market, factor, financing, and technology exposures. A risk-taking construction deliberately accepts a larger exposure budget in pursuit of a larger payoff if the thesis is correct. Neither label implies that the portfolio is safe or that its expected return is positive.

1.1 Notation

Let N, A, and G denote NVIDIA, Amazon, and Google. Let w_N be the portfolio weight in NVIDIA and w_A, w_G the weights in Amazon and Google. A long position is positive and a short position is negative.

The core NAGMO sign restrictions are

$$w_N > 0, \qquad w_A < 0, \qquad w_G < 0. \tag{3}$$

Meta remains excluded from the core constructions in order to preserve the original N–A–G hypothesis. It may be introduced only as an explicitly justified additional hedge or factor-control instrument, not silently added to the thesis.

Define net risky exposure and gross exposure by

$$N_w = w_N + w_A + w_G, \tag{4}$$
$$G_w = |w_N| + |w_A| + |w_G|. \tag{5}$$

Dollar neutrality is $N_w = 0$. It is not the same as beta neutrality:

$$\beta_p = \beta_N w_N + \beta_A w_A + \beta_G w_G. \tag{6}$$

The portfolio's realized net return, before investor-specific tax, is represented as

$$R_p = w_N r_N + w_A r_A + w_G r_G - C_{\text{fin}} - C_{\text{borrow}} - C_{\text{trade}}, \tag{7}$$

where the cost terms include financing, stock-loan, and execution costs. Cash collateral, margin requirements, and short-sale proceeds must be handled consistently in an implementation ledger.

2 The risk-averse family

2.1 Construction I: Low-gross, dollar-neutral NAGMO

The first defensive construction retains the original direction of the thesis but reduces the gross exposure. Set a positive long notional L, and allocate the short book between Amazon and Google using a split parameter λ, where $0 \le \lambda \le 1$:

$$w_N = L, \tag{8}$$
$$w_A = -\lambda L, \tag{9}$$
$$w_G = -(1-\lambda)L. \tag{10}$$

Then

$$N_w = L - \lambda L - (1-\lambda)L = 0, \qquad G_w = 2L. \tag{11}$$

This construction makes the short allocation transparent. An equal split uses $\lambda = 1/2$; a different split requires an explicit reason, such as relative volatility, valuation, factor loading, or the strength of the cloud-specific hedge. The value of L is determined by the portfolio's gross-exposure budget, not by the market capitalizations in the original arithmetic identity.

This is the simplest risk-averse implementation when the investor wants to preserve the NAGMO spread while limiting position scale. It does not guarantee beta neutrality. Its beta is

$$\beta_p = L\left[\beta_N - \lambda\beta_A - (1-\lambda)\beta_G\right]. \tag{12}$$

Accordingly, the split parameter should not be selected solely to make the short notionals look symmetrical.

2.2 Construction II: Constrained beta-neutral spread

A second defensive construction solves for the short allocation that offsets the estimated market beta of the NVIDIA long. For a given L, impose

$$w_N = L, \tag{13}$$
$$w_A = -\lambda L, \qquad w_G = -(1-\lambda)L, \tag{14}$$
$$\beta_p = 0. \tag{15}$$

The beta-neutral split, when it exists within the admissible interval, is

$$\lambda^* = \frac{\beta_N - \beta_G}{\beta_A - \beta_G}, \qquad \beta_A \neq \beta_G. \tag{16}$$

A feasible solution requires $0 \leq \lambda^* \leq 1$. If the solution lies outside this interval, the core NAGMO short legs cannot achieve beta neutrality under the estimated betas and the sign restrictions. The correct response is not to conceal the residual; it is to report it, reduce L, or consider a separately disclosed hedge instrument.

Because betas are estimated rather than known, exact in-sample neutrality can be fragile. A more robust version minimizes expected squared beta across an estimation uncertainty set:

$$\min_{\lambda \in [0,1]} \mathbb{E}_\theta \left[\left(\beta_N(\theta) - \lambda \beta_A(\theta) - (1-\lambda)\beta_G(\theta) \right)^2 \right], \tag{17}$$

where θ indexes plausible estimation states. The objective should be evaluated out of sample and across multiple return windows.

2.3 Construction III: Multi-factor and AI-risk-constrained portfolio

Market beta is only one source of systematic exposure. A defensive portfolio can impose limits on a vector of factor exposures:

$$\mathbf{f}_p = \mathbf{B}^\mathsf{T}\mathbf{w}, \qquad \mathbf{w} = (w_N, w_A, w_G)^\mathsf{T}, \tag{18}$$

where $\mathbf{B}$ contains estimated factor loadings. Factors may include broad equity market, momentum, growth, quality, size, volatility, rates, and technology-sector exposure. The portfolio can minimize a covariance-based risk measure subject to the NAGMO sign restrictions and exposure limits:

$$\min_{\mathbf{w}} \quad \mathbf{w}^\mathsf{T}\boldsymbol{\Sigma}\mathbf{w} + \eta G_w + \kappa C(\mathbf{w}) \tag{19}$$

$$\text{subject to} \quad w_N > 0, \quad w_A < 0, \quad w_G < 0, \tag{20}$$

$$|N_w| \leq \epsilon_N, \quad |\mathbf{f}_p| \leq \boldsymbol{\epsilon}_f, \quad G_w \leq G_{\max}. \tag{21}$$

Here $\boldsymbol{\Sigma}$ is the estimated covariance matrix, $C(\mathbf{w})$ is an implementation-cost estimate, and η, κ are penalty parameters. The tolerances are investment-policy choices, not universal constants.

AI model risk should be represented as explicit scenarios rather than assumed to be captured by historical covariance. Let s index scenarios such as compute scarcity, compute abundance, custom-silicon substitution, model commoditization, and weak AI monetization. Let $\mathbf{q}_s$ be the estimated security-return shock under scenario s. A defensive construction may impose

$$\mathbf{w}^\mathsf{T}\mathbf{q}_s \geq -D_s \qquad \text{for every scenario } s, \tag{22}$$

where D_s is the maximum tolerated loss under that scenario. Scenario shocks are assumptions to be documented and challenged; they are not forecasts.

2.4 Construction IV: Inflation-risk-premium-aware defensive portfolio

The defensive portfolio should not rely on a single assumed sign of the U.S. inflation risk premium. Let ϕ denote the premium and Φ an admissible set of plausible values. The stylized beta relation in the Conditional Impartiality Theorem is

$$\beta_p(\phi) = \beta_{p,0} + (\delta_N - \delta_A - \delta_G)(\phi - \phi_0), \tag{23}$$

under its illustrative linear specification. The numerical crossing near 0.50% in that paper is a calibration result, not a live estimate or a universal threshold.

A robust defensive objective can minimize the worst estimated risk across Φ:

$$\min_{\mathbf{w}} \max_{\phi \in \Phi} \left\{ \mathbf{w}^\mathsf{T}\boldsymbol{\Sigma}(\phi)\mathbf{w} + \rho_\beta \beta_p(\phi)^2 + \rho_c C(\mathbf{w}, \phi) \right\}. \tag{24}$$

This formulation does not assume that a negative ϕ automatically creates favorable carry or eliminates beta risk. It requires the portfolio to remain within its risk budget across positive, near-zero, and negative premium scenarios. Since ϕ is latent and estimated with uncertainty, Φ should be an interval or probability distribution supported by a stated estimation method.

3 The risk-taking family

Risk-taking constructions preserve the same underlying relative-value thesis but use a larger risk budget, a more concentrated short allocation, or a deliberate tolerance for residual factor exposure. The distinction is in the size and governance of the risk budget, not in a claim that risk has disappeared.

3.1 Construction V: Conviction-weighted NAGMO spread

A conviction-weighted portfolio allocates the short book according to a documented relative-value assessment:

$$w_N = L, \tag{25}$$

$$w_A = -\lambda L, \qquad w_G = -(1-\lambda)L, \tag{26}$$

but permits a larger L and a λ chosen from a forecast of relative returns rather than solely from beta neutrality. Let $\widehat{\boldsymbol{\mu}}$ be the vector of expected security returns. A mean-variance formulation is

$$\max_{\mathbf{w}} \quad \widehat{\boldsymbol{\mu}}^{\mathsf{T}}\mathbf{w} - \frac{\gamma}{2}\mathbf{w}^{\mathsf{T}}\boldsymbol{\Sigma}\mathbf{w} - \kappa C(\mathbf{w}), \tag{27}$$

subject to the NAGMO sign restrictions and a stated gross-exposure ceiling. A lower γ represents a greater willingness to accept estimated variance for expected return. It does not justify ignoring model uncertainty or tail loss.

The expected-return vector must be a relative-return forecast, not simply a bullish forecast for NVIDIA. If all three securities rise, the trade can still lose; if all three fall, it can still win. The relevant object is the return spread after costs.

3.2 Construction VI: Deliberate residual-beta portfolio

A risk-taking investor may choose not to neutralize beta fully. This can be appropriate only if the residual exposure is deliberate, quantified, and within a pre-agreed risk budget. Define a target beta $\bar{\beta}$, which may be positive or negative:

$$\beta_p \approx \bar{\beta}. \tag{28}$$

The portfolio is then optimized for expected relative return subject to

$$|\beta_p - \bar{\beta}| \le \epsilon_\beta, \qquad G_w \le G_{\max}. \tag{29}$$

This should be described as a beta-budgeted NAGMO portfolio, not as market-neutral. Its performance attribution must separate the relative-value component from the return generated by the intentional market exposure.

The target beta should be tested under alternative ϕ regimes. If the sign of the residual beta changes as ϕ changes, the investor must decide whether that is acceptable, whether the position should be scaled down, or whether a separate hedge is warranted. A fixed target based on one historical calibration is not robust by construction.

3.3 Construction VII: High-conviction, capped-gross portfolio

A high-conviction construction increases the long and short notionals while imposing hard limits on gross exposure, single-name concentration, and scenario loss. Its risk budget can be expressed as

$$G_w \le G_{\max}, \tag{30}$$

$$|w_i| \le w_{\max} \quad \text{for each security } i, \tag{31}$$

$$\mathbf{w}^{\mathsf{T}}\mathbf{q}_s \ge -D_s \quad \text{for all designated stress scenarios } s. \tag{32}$$

A CVaR-based alternative controls the expected loss in the tail of the return distribution:

$$\min_{\mathbf{w}} \quad -\widehat{\boldsymbol{\mu}}^{\mathsf{T}}\mathbf{w} + \gamma\ \mathrm{CVaR}_\alpha(-R_p) + \kappa C(\mathbf{w}). \tag{33}$$

The confidence level α, stress horizon, and loss budget must be stated in advance. Historical tail estimates alone may understate losses from a sudden technology or policy discontinuity.

3.4 Construction VIII: Conditional, regime-switching NAGMO

The most adaptive risk-taking construction changes its exposure scale as the joint AI and monetary regime changes. Let z_t represent the AI regime and ϕ_t the estimated U.S. inflation risk premium. Define a scaling function s_t bounded by zero and one:

$$0 \leq s_t = s(z_t, \phi_t, \mathcal{R}_t) \leq 1, \tag{34}$$

where $\mathcal{R}_t$ contains measured risk indicators such as volatility, liquidity, borrow availability, and factor instability. The portfolio weights are

$$\mathbf{w}_t = s_t \mathbf{w}_t^*, \tag{35}$$

where $\mathbf{w}_t^*$ is the target construction under the current estimates. The scaling rule should reduce exposure when the uncertainty of the AI thesis or the uncertainty of ϕ_t rises, when borrow becomes unreliable, or when stress losses exceed the risk budget.

A regime-switching rule should not mechanically increase leverage merely because a point estimate of ϕ crosses the stylized 0.50% threshold. The threshold belongs to the illustrative calibration in the Conditional Impartiality Theorem. A live implementation requires re-estimation, confidence intervals, and an explicit decision rule.

4 Comparison of the constructions

Construction	Primary objective	Main control	Principal residual risk
I. Low-gross	Preserve spread at modest scale	Gross-exposure cap	Residual beta and factor exposure
II. Beta-neutral	Reduce market sensitivity	Estimated beta constraint	Beta estimation error and other factors
III. Multi-factor	Limit systematic risk	Factor and scenario constraints	Model misspecification and tail events
IV. IRP-aware	Robustness across monetary regimes	Worst-case ϕ objective	Uncertainty in estimating ϕ
V. Conviction-weighted	Maximize forecast spread return	Risk-adjusted sizing	Forecast and valuation error
VI. Residual-beta	Accept a declared beta budget	Target-beta constraint	Broad market direction
VII. High-conviction	Increase exposure within hard limits	Gross, name, and CVaR caps	Gap, squeeze, and technology shocks
VIII. Regime-switching	Adapt exposure to changing states	Dynamic scaling rule	Regime-classification and timing error

No construction dominates in every state. A low-gross portfolio may be preferable when the thesis is plausible but evidence is weak. A beta-neutral construction is appropriate when market exposure is not part of the intended bet. A multi-factor or IRP-aware construction is preferable when the investor explicitly wants the result to survive changes in macroeconomic conditions. A conviction-weighted or high-gross construction is suitable only when the investor has a defensible relative-return estimate and the capacity to withstand losses. A regime-switching design may help manage changing conditions, but introduces estimation, turnover, and timing risks of its own.

5 Common stress-testing protocol

All eight constructions should be tested against the same set of scenarios. The scenario design should distinguish shocks to AI economics from shocks to monetary conditions.

1. **Compute scarcity:** demand for advanced accelerators remains strong, utilization is high, and supply constraints support hardware pricing power.
2. **Compute abundance:** capacity expands, inference costs fall, and price competition compresses unit economics.

3. **Custom-silicon substitution:** cloud providers move selected workloads to internally designed or alternative accelerators.

4. **Model commoditization:** open-weight and lower-cost models reduce scarcity rents while potentially expanding aggregate usage.

5. **Weak monetization:** AI adoption grows, but revenue conversion and margins disappoint.

6. **Positive, zero, and negative ϕ:** nominal-rate and financing assumptions are re-estimated under each premium regime; no sign is treated as permanent.

7. **Liquidity and short squeeze:** borrow fees rise, shares become difficult to source, margin requirements increase, or a short leg gaps upward.

8. **Joint shock:** an AI-specific adverse event occurs alongside a material change in real yields, expected inflation, or the inflation risk premium.

For each scenario, report portfolio return, maximum drawdown, beta, factor exposures, gross exposure, financing and borrow costs, liquidity-adjusted exit cost, and the contribution of each leg. Report both the point estimate and uncertainty range. The analysis should use synchronized total-return data, avoid look-ahead bias, include dividends and borrow costs, and employ walk-forward validation. It should compare NAGMO with a long-only NVIDIA position, a simple equal-notional spread, and a properly estimated beta-neutral alternative.

6 Implementation governance and selection rule

Before capital is committed, the investment process should record: (i) the thesis horizon; (ii) the maximum gross and net exposures; (iii) the permitted beta and factor budgets; (iv) the source and uncertainty of the ϕ estimate; (v) the AI scenarios and loss limits; (vi) borrow availability and financing terms; and (vii) the conditions for reducing or closing the trade.

A practical selection rule is sequential. Begin with Construction I at a small risk budget. Move to Construction II or III only when reliable estimates support the additional constraints. Use Construction IV when the investment mandate requires robustness to monetary-regime uncertainty. Adopt Constructions V–VII only when a documented expected-return advantage survives cost and stress analysis. Use Construction VIII only when the regime indicators are observable, the scaling rule is specified ex ante, and turnover costs are included.

The investor should not select a risk-taking architecture merely because the risk-averse one has a lower expected return in a point forecast. Nor should a risk-averse architecture be called safe merely because it has lower gross exposure. Concentrated technology risk, short squeezes, model discontinuities, and estimation error can remain material at small position sizes.

7 Conclusion

The NAGMO thesis can be implemented through multiple portfolio architectures, each expressing a different tolerance for market exposure, factor risk, AI model uncertainty, financing costs, and monetary-regime uncertainty. The risk-averse family emphasizes low gross exposure, beta and factor controls, scenario limits, and robustness across plausible values of the U.S. inflation risk premium. The risk-taking family accepts a larger risk budget through conviction weighting, deliberate residual beta, capped high-conviction exposure, or regime-dependent scaling.

The central methodological requirement remains unchanged: do not confuse dollar balance with market neutrality, and do not treat any nominal-rate critique as regime-independent when it depends on ϕ. Equally, a change in the sign of ϕ does not remove AI model risk or guarantee favorable carry. The stylized inversion threshold in the Conditional Impartiality Theorem is a useful prompt for conditional analysis, not a substitute for current estimation.

The preferred construction is therefore not a universal set of weights. It is the architecture whose constraints match the investor's risk mandate and whose estimated exposures remain acceptable across AI scenarios, financing conditions, and positive, near-zero, and negative inflation-risk-premium regimes. Until the necessary return, covariance, borrow, cost, and scenario data are supplied, all weight choices should remain parameterized or explicitly illustrative rather than presented as empirically optimal.

References

[1] Ghosh, S. (2025). *The NAGMO Investment Thesis: A Market-Neutral Portfolio in AI*. Kolkata, India.

[2] Ghosh, S., with Z.ai. (2026). *The Conditional Impartiality Theorem: On the Hidden Dependence of Every NAGMO Critique on the Sign of the US Inflation Risk Premium*. Kolkata, India.

[3] Ghosh, S. (2026). *Review of the NAGMO Investment Thesis: Market Neutrality, AI Model Risk, and the Conditional Role of the U.S. Inflation Risk Premium*. Kolkata, India.

[4] Fisher, I. (1930). *The Theory of Interest*. New York: Macmillan.

[5] Campbell, J. Y., and Viceira, L. M. (2002). *Strategic Asset Allocation: Portfolio Choice for Long-Term Investors*. Oxford: Oxford University Press.

[6] Ilmanen, A. (2011). *Expected Returns: An Investor's Guide to Harvesting Market Rewards*. Hoboken: Wiley.

[7] Grinold, R. C., and Kahn, R. N. (2000). *Active Portfolio Management: A Quantitative Approach for Providing Superior Returns and Controlling Risk*. New York: McGraw-Hill.

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